

HARMFUL EFFECTS OF REPRESENTATIVENESS BIAS

Examples of the Harmful Effects of Sample-Size Neglect for Investors

1. Investors can make significant financial errors when they examine a money manager's track record. They peruse the past few quarters or even years and conclude, based on inadequate statistical data, that the fund's performance is the result of skilled allocation and/or security selection.
2. Investors also make similar mistakes when investigating track records of stock analysts. For example, they look at the success of an analyst's past few recommendations, erroneously assessing the analyst's aptitude based on this limited data sample.

Examples of the Harmful Effects of Base-Rate Neglect for Investors

1. What is the probability that Company A (ABC, a 75-year-old steel manufacturer that is having some business difficulties) belongs to group B (value stocks that will likely recover) rather than to Group C (companies that will go out of business)? In answering this question, most investors will try to judge the degree to which A is representative of B or C. In this case, some headlines featuring recent bankruptcies by steel companies make ABC Steel appear more representative of the latter categorization, and some investors conclude that they had best unload the stock. They are ignoring, however, the base-rate reality that far more steel companies survive or get acquired than go out of business.
2. What is the probability that AAA-rated Municipal Bond A (issued by an "inner city" and racially divided county) belongs to Group B (risky municipal bonds) rather than to Group C (safe municipal bonds)? In answering this question, most investors will again try to evaluate the extent to which A seems representative of B or C. In this case, Bond A's characteristics may seem representative of Group A (risky bonds) because of the county's "unsafe" reputation; however, this conclusion ignores the base-rate fact that, historically, the default rate of AAA bonds is virtually zero.